

Reconstruct from a finite sample set

Background

Ideal interpolation uses $x_r(t) = \sum x[n] \text{sinc}(fs(t-t[n]))$, where $\text{sinc}(u) = \sin(\pi u)/(\pi u)$, with $\text{sinc}(0)=1$. The theoretical sum is infinite. Here use only the supplied samples: the finite reconstruction still matches them at their sample instants but need not equal the original signal between samples.

Your task

Reconstruct $x(t) = \cos(2\pi 3t)$ from samples with indices $-M$ through M . Evaluate the reconstruction on the supplied dense time grid. Report both the maximum error at the sample instants and the RMSE on the dense grid. RMSE means the square root of the mean squared difference.

Function contract and assumptions

`reconstruct(t_eval, t_samples, samples, fs)` returns a one-dimensional array matching `t_eval`. `rmse(reference, estimate)` returns a scalar. The provided `t_samples` are distinct integer multiples of $1/fs$. You may use the normalized built-in `sinc`. Do not use ZOH or linear interpolation.

Example / correctness check

With `t_samples=[-0.25,0,0.25]`, `samples=[0,1,0]`, `fs=4` and `t_eval=[0,0.125,0.25]`, the output is approximately `[1,0.63661977,0]`. Any supplied sample values should be reproduced at their own sample times to roundoff.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF page 27; finite-window error and RMSE are defined above.